

Galvanic Corrosion + Rust Removal

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Corrosion, sacrificial protection, and reversing rust

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 5 — Galvanic Corrosion and Electrolytic Rust Removal

Hook: Connect corrosion, sacrificial protection, and reversing rust with electricity.

Learning outcomes

- Build metal-pair galvanic cells and measure voltage
- Explain galvanic corrosion and sacrificial protection
- Interpret overnight electrolytic rust removal (started Session 4)
- Synthesize the week's experiments in a final review

Session plan (90 min)

Time	Activity	Reference
0–15 min	Hook: reveal derusted nails + corrosion jars	experiment.md — Part C
15–30 min	Galvanic cells, corrosion, sacrificial anodes	lecture.md
30–50 min	Metal-pair cells — measure voltage	experiment.md — Part A
50–60 min	Corrosion jar observations	experiment.md — Part B
60–75 min	Week synthesis: spontaneous vs driven cells	lecture.md
75–90 min	Final challenge questions	lecture.md

Key reactions

Galvanic: $\text{Zn(s)} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

Electrolytic derusting (cathode): rust reduced toward Fe; H_2 may evolve

Sacrificial anode (+): metal oxidizes

Multi-day setup

Demo	Start	Reveal
Corrosion jars A/B/C	Session 1–2	Session 5
Electrolytic rust removal	Session 4 (end)	Session 5 (opening)

Status

- [] Rust-removal cells started Session 4 end-of-class
- [] Metal samples prepared (Zn, Fe, Cu, Al)
- [] Final review questions prepared

- ☐ Washing soda or baking soda procured

Session 5 — Preparation

Before class (instructor)

- [] Confirm derusting cells were started Session 4 end-of-class (see Session 4 prep)
- [] Photograph nails before treatment if not done Thursday
- [] Prepare saltwater for galvanic demos (label "**corrosion only — not for gas electrolysis**")
- [] Clean metal strips; sand if oxidized
- [] Print final challenge worksheets

Session 4 handoff (critical)

Derusting cells must be running **before students leave Session 4**:

Item	Detail
Electrolyte	Washing soda preferred; baking soda OK — not NaCl
Cathode (–)	Rusty iron
Anode (+)	Sacrificial steel or graphite
Control	Rusty nail in plain water, no power
Power	9 V battery; secure clips; label each cell

Optional multi-day corrosion setup

Start Monday or Tuesday for best rust by Friday:

Jar	Setup
B	Iron nail wired/touching copper in saltwater
C	Iron nail wired/touching zinc in saltwater
A	Iron alone (control)

Store open jars where students can observe daily changes.

Day-of timeline

When	Task
0 min	Reveal derusted nails + corrosion jars (Part C)
25 min	Metal-pair galvanic cells (Part A)
45 min	Corrosion jar observations (Part B)

When	Task
70 min	Week comparison table (Part D)
80 min	Final challenge; collect worksheets

Rust removal troubleshooting

Issue	Fix
Little visible change overnight	Fresher rust; higher soda concentration; check connections
Anode heavily corroded	Expected — sacrificial behavior
Cloudy solution	Normal; rinse nail before comparing
Chlorine smell	Stop — salt was used; replace electrolyte

Open questions / notes

- Washing soda vs. baking soda used:
- Overnight run time (hours):
- Corrosion jars started on date:

Session 5 — Materials

Part C — Electrolytic rust removal (started Session 4)

- ☐ Rusty iron nails or small rusty hardware (2+ per group)
- ☐ Washing soda: **1 tbsp / ~250 mL** water (or baking soda same)
- ☐ Sacrificial steel strips or graphite rods (anodes)
- ☐ Small plastic cups or jars
- ☐ 9 V batteries and alligator clips
- ☐ Control container (plain water, no power)
- ☐ Camera or labels for before/after comparison

Part A & B — Galvanic corrosion

Per group

- ☐ Saltwater **5–10% w/v** (25–50 g NaCl per 500 mL) — corrosion demos only, labeled
- ☐ Copper strip or wire
- ☐ Iron nail
- ☐ Zinc strip or galvanized nail
- ☐ Aluminum foil
- ☐ Multimeter
- ☐ Alligator clips
- ☐ Small jars (3 for comparison set)
- ☐ Paper towels

Optional

- ☐ Magnesium ribbon

Instructor

- ☐ Photos of sacrificial anodes (ships, water heaters)
- ☐ Final review worksheets

Multi-day prep

- ☐ Derusting cells started **Session 4** end-of-class
- ☐ Jars B and C set up at start of week for visible corrosion (optional)

Session 5 — Experiment: Corrosion and Electrolytic Rust Removal

Part C opens the session (overnight reveal). **Parts A & B** are student hands-on.

Instructors: jar narratives, metal-pair voltages, reveal protocol — instructor.md.

Solutions used today — concentrations

Derusting bath (started Session 4 — for reveal)

Ingredient	Amount
Washing soda (Na_2CO_3) preferred	1 tbsp (~15 g) per ~250 mL warm water
or baking soda (NaHCO_3)	1 tbsp per ~250 mL warm water

No NaCl. Tap water is fine for derusting baths.

Saltwater for galvanic / corrosion demos only

Ingredient	Amount
Table salt (NaCl)	25–50 g per 500 mL water
Water	500 mL tap water

≈ **5–10% w/v NaCl** ("strong saltwater"). Tap water is fine.

Label clearly: "**CORROSION / GALVANIC ONLY — NOT for electrolysis or derusting**".

Part C — Electrolytic rust removal reveal (0–15 min)

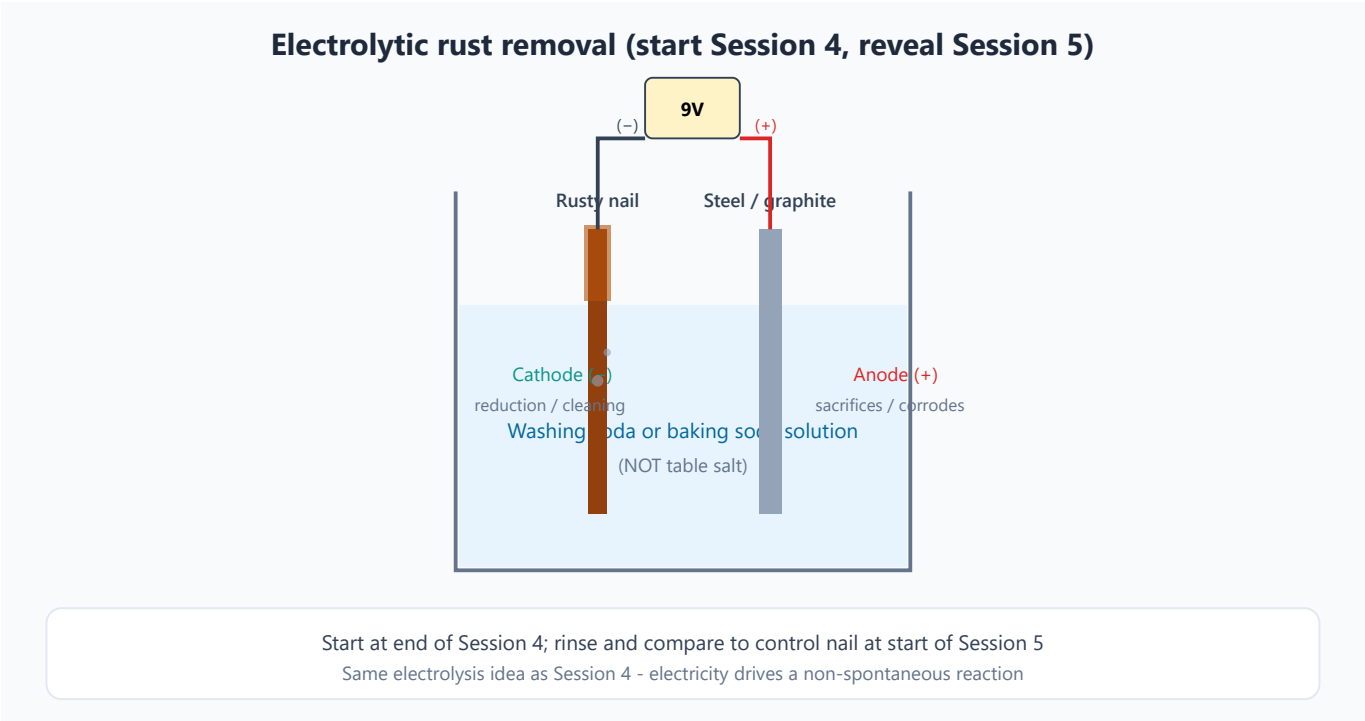


Figure 1 — Cells started at end of Session 4: rusty iron on cathode (–), sacrificial anode on (+), in washing soda or baking soda solution.

Reminder of overnight setup

- 1. Jar with washing soda or baking soda solution (recipe above).
- 2. **Cathode (–)**: rusty iron nail or hardware.
- 3. **Anode (+)**: sacrificial steel strip or graphite.
- 4. 9 V battery left running overnight (or until Session 5).
- 5. Control: rusty nail in plain water, no power.

Today's reveal procedure

- 1. **Disconnect** the battery before touching electrodes.
- 2. Remove the treated nail with forceps or gloves; rinse under water.
- 3. Compare side-by-side with the control and the before photo.
- 4. Note: cathode bubbles (H₂ possible), anode darkening/sludge, bath cloudiness.
- 5. Discuss spontaneous rust (jars) vs driven cleaning (this cell).

Observation sheet

Sample	Electrolyte	Power overnight?	Appearance after rinse	Cleaner than control?
Treated nail	Washing soda / baking soda	Yes		

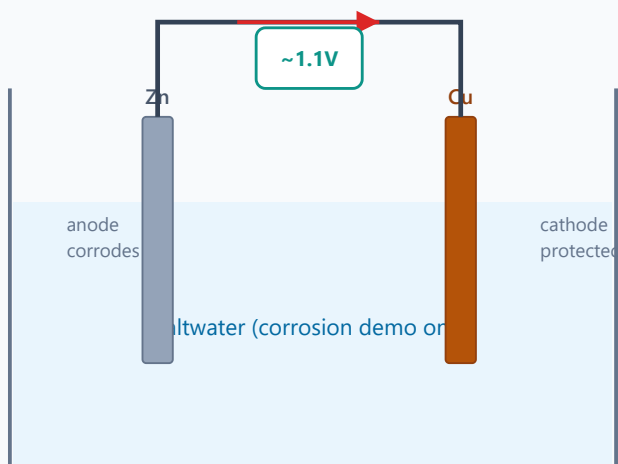
Sample	Electrolyte	Power overnight?	Appearance after rinse	Cleaner than control?
Control nail	Water only	No		
Sacrificial anode	—	Yes		

Discussion prompts

- Which electrode was the **cathode**? What happens there?
- How is this like Session 4 electrolysis?
- How is this **opposite** of jar B (Fe + Cu) corrosion?

Part A — Metal-pair galvanic cells (25–45 min)

Session 5 - Galvanic cell (Zn + Cu in saltwater)



$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (oxidation at anode) | Spontaneous - produces electricity like Session 1
More active metal becomes anode; less active metal becomes cathode

Figure 2 — Spontaneous cell: Zn corrodes (anode), Cu is cathode; measure ~1.1 V with a multimeter.

Setup

1. Pour **~50–100 mL** of **5–10% saltwater** into a small jar.
2. Clean metal strips lightly with sandpaper if heavily oxidized.
3. For each pair: immerse both metals so they do **not touch** in solution.
4. Multimeter on DC volts: probe one metal with red, the other with black.
5. Record voltage and which metal is at the **+** meter terminal.

Pairs to test

Pair	Predicted anode (corrodes)	Typical voltage (order of magnitude)
Zn + Cu	Zn	~0.9–1.1 V
Fe + Cu	Fe	~0.5–0.8 V
Al + Cu	Al (may be suppressed by oxide)	variable
Zn + Fe	Zn	lower than Zn–Cu

Procedure tips

- Keep immersion depth similar across trials.
- If voltage is near zero, check that metals are not shorted and that saltwater is strong enough.
- Wipe electrodes between trials to reduce cross-contamination.

Data table

Metal 1	Metal 2	V (V)	(+) terminal on	Anode (corroding)
Zn	Cu			
Fe	Cu			
Al	Cu			
Zn	Fe			

Part B — Corrosion comparisons (45–55 min)

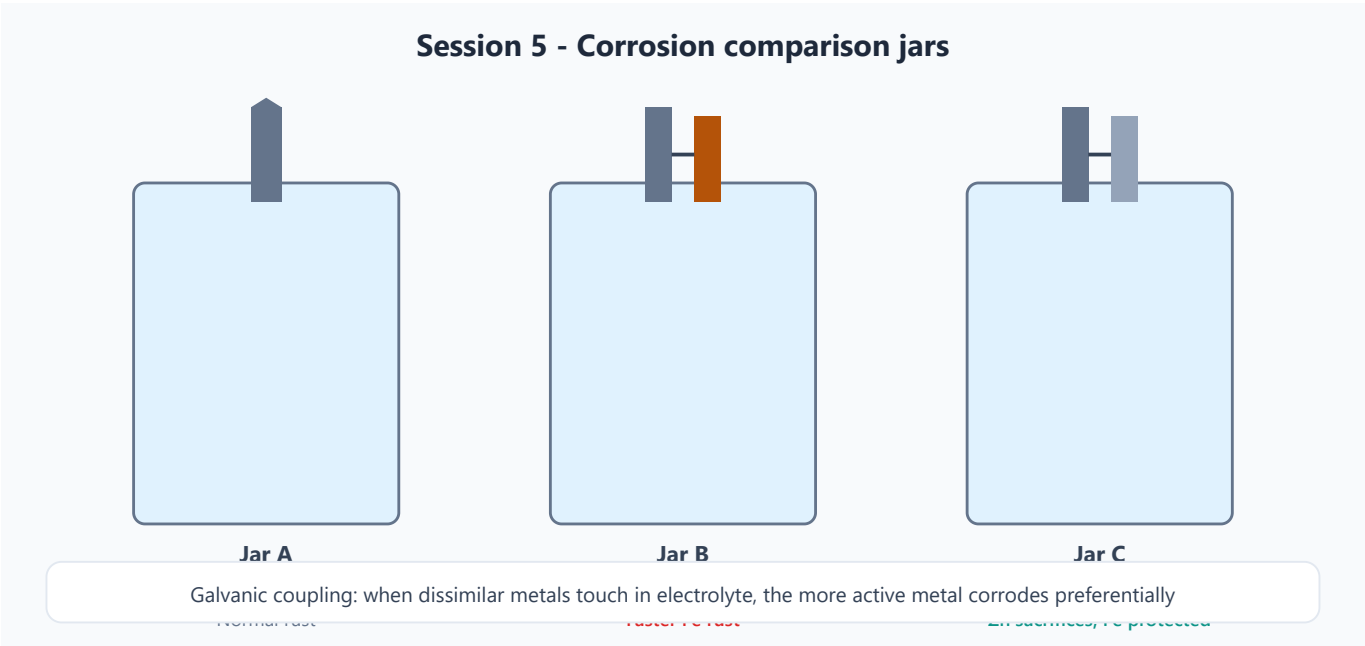


Figure 3 — Jar B: iron rusts faster when coupled to copper. Jar C: zinc sacrificially protects iron.

Jar recipes (set up earlier in the week if possible)

Use the same **5–10% saltwater** in each jar (~100–150 mL per jar).

Jar	Contents	Expected over days
A	Iron nail alone	Gradual rust
B	Iron + copper wire/strip touching the iron	Faster iron rust
C	Iron + zinc (or galvanized nail) touching the iron	Iron protected; zinc corrodes

Contact matters: metals must touch (or be clipped together) so electrons can flow.

Today's task

1. Observe jars A/B/C (multi-day or same-day start).
2. Compare rust color, cloudiness, bubbles, metal appearance.
3. Explain with the galvanic model (activity series).
4. Link to Part C: sacrificial anode idea (Zn in jar C vs steel anode in derusting).

Observation sheet

Jar	Iron appearance	Partner metal	Galvanic couple?	Which corrodes?
A		—		
B		Cu		
C		Zn		

Part D — Compare operations (70–80 min)

Feature	Fruit battery	Electroplating	Water electrolysis	Rust removal
Needs external power?				
Produces electricity?				
Gas involved?				
Metal transferred?				
Reverses corrosion?				

Final challenge worksheet

See lecture.md — Final challenge questions.

Safety

- Saltwater corrosion demos: wash hands after; optional gloves
 - Keep saltwater **away** from any leftover Session 4 electrolysis tubs
 - Derusting: washing soda/baking soda only — **no NaCl**
 - Disconnect powered cells when the room is unsupervised
 - No open flames near gas-evolving setups
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Experiment status

- ☐ Derusting cells started Session 4 with documented electrolyte recipe
- ☐ Control nail and before photo ready
- ☐ 5–10% saltwater mixed and labeled for galvanic/corrosion only
- ☐ Corrosion jars prepared (same-day or multi-day)
- ☐ Metal strips cleaned and labeled
- ☐ Final worksheet printed

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution	No saltwater; no flames near setup; instructor-only pop test
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H₂/O₂ container